



Build a Virtual Private Cloud (VPC)

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The screenshot displays the AWS Management Console interface for creating a new Virtual Private Cloud (VPC). The breadcrumb navigation at the top indicates the path: **VPC** > **Your VPCs** > **Create VPC**. The main heading is **Create VPC** with an **Info** link. A descriptive text states: "A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances."

VPC settings

Resources to create **Info**
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

NextWork VPC

IPv4 CIDR block **Info**
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block **Info**
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me



Introducing Today's Project!

In this project, I will demonstrate my skill in building a VPC using ASW console...
I'm doing this project to learn and develop my cloud infrastructure knowledge ...

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is an isolated virtual network inside a public cloud environment. It allows me my very own section of a cloud where I am able to launch and organise resources like EC2 instances that are separate from everyone else's. It is useful because without it there would be no network boundaries between different users' resources, making it impossible to control access or keep environments secure.

In today's project, I used Amazon VPC to launch a subnet within it and attach an internet gateway to that subnet to allow the VPC to have internet access.

Personal reflection

This project took me roughly 1 hour to complete.



One thing I didn't expect in this project was having the learning path be interactive as opposed to tutorial-like where no questions are asked and I am not really tested. In this project I was able to grasp concepts much better because of how my understanding of each topic was questioned along with practical elements of using the console. I am looking forward to using the AWS CLI in future projects!



Virtual Private Clouds (VPCs)

What I did in this step

In this step, I will access the VPC console in AWS Portal to create a VPC.

How VPCs work

VPCs are isolated virtual networks in a public cloud environment. It gives you your own private section of the cloud where you can launch and organize resources separate from everyone else's. Its like your own fenced-off section in a massive shared data centre. Without VPC's, there would be no network boundaries between different user's resources, making it impossible to keep environments and personal data secure.

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This is because creates ne for every new account. They have done this to make it easier for new users to get started - without it,beginners would have to learn how to create and configure a VPC before they could launch basic resources like EC2 instances. It is worth noting that not all AWS services require a VPC. There are other resources such as Lambda that can run without one. However, resources like EC2 and databases require a secure private network to communicate with each other.



The screenshot shows the AWS Management Console interface for creating a new VPC. The breadcrumb navigation at the top indicates the path: VPC > Your VPCs > Create VPC. The main heading is 'Create VPC' with an 'Info' link. A descriptive sentence states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.'

The 'VPC settings' section contains the following configuration options:

- Resources to create:** Two radio button options are present: 'VPC only' (which is selected) and 'VPC and more'.
- Name tag - optional:** A text input field contains the value 'NextWork VPC'. A note below states: 'Creates a tag with a key of 'Name' and a value that you specify.'
- IPv4 CIDR block:** Two radio button options are present: 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block'.
- IPv4 CIDR:** A text input field contains the value '10.0.0.0/16'. A note below states: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block:** Four radio button options are present: 'No IPv6 CIDR block' (selected), 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me'.

Defining IPv4 CIDR blocks

An IPv4 CIDR block is a range of IP addresses defined using IPv4 format and CIDR(Classless Inter-Domain Routing). It allows you to allocate a whole block of addresses at once rather than one by one. For example, 10.0.0.0/16 means the first 16 bits (the 10.0 portion) are fixed, and the remaining bits are available. This gives us 65,536 possible addresses. The number after the slash is the prefix length: the smaller it is, the larger the block of addresses we get.



Subnets

What I did in this step

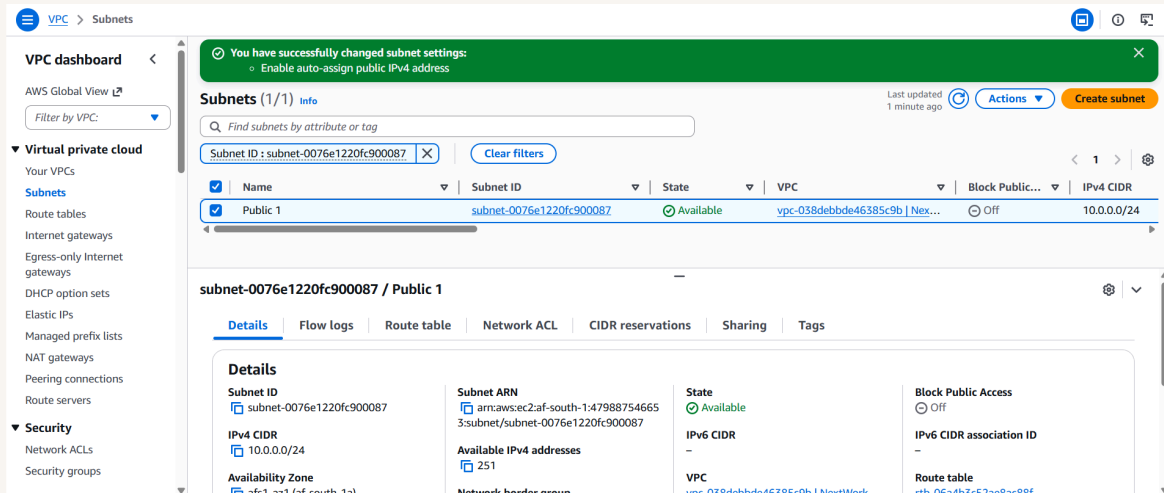
In this step, I will divide my VPC into subdivisions by creating subnets so that I can start planning where different resources will run and operate.

Creating and configuring subnets

Subnets are smaller segments of a VPC's IP address range, used to organise resources within the network. There are two types of subnets: public subnets, which have a route to an internet gateway so that resources inside them can access the internet, and private subnets, which have no internet route and are used for resources like databases that shouldn't be publicly exposed. My account already has default subnets. One for each Availability Zone(AZ) in the region. An AZ is a cluster of data centres within a region, and each subnet must belong to a specific AZ. By spreading resources across multiple AZ's we gain high availability. If one AZ goes down, the others keep our applications running without disruption.

Public vs private subnets

The difference between public and private subnets are that public subnets are connected to the internet and can communicate with external networks. Private subnets do not have direct access to the internet and are used for internal resources that do not need to be publicly accessible ... For a subnet to be considered public, it has to be connected to an internet gateway.



Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled auto-assign public IPv4 address. This setting makes sure that my subnet has a public IP address so that when I launch any EC2 instance, I won't have to create one manually.

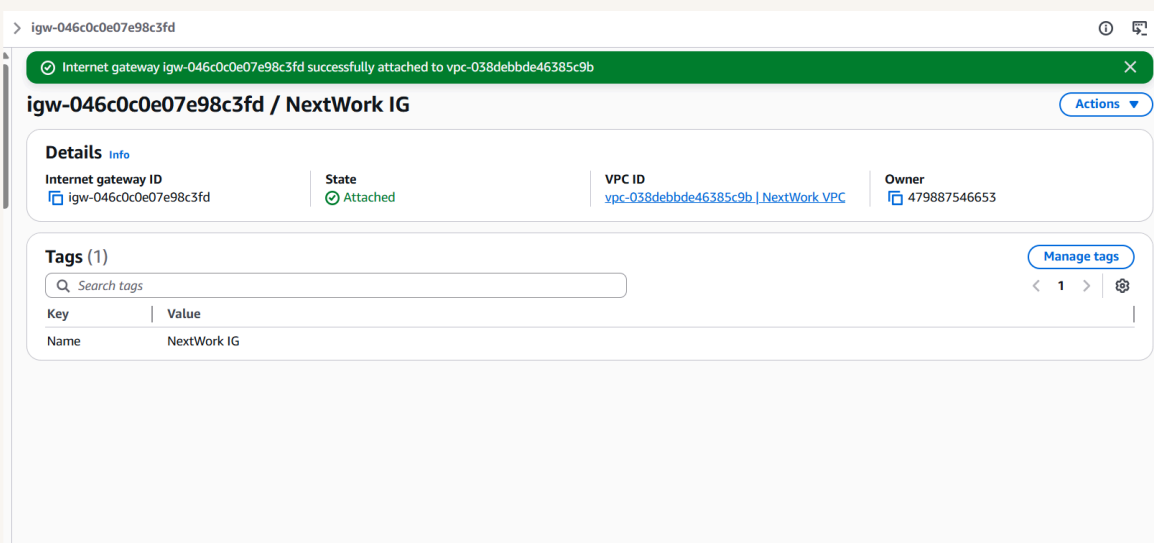


Internet gateways

Setting up internet gateways

Internet gateways are cloud components that connects your VPC to the internet and be accessible to external users.It allows resources with public IP addresses to send traffic out to the internet and receive traffic from the external users. Also, it's what determines whther a subnet is public or private. As mentioned before, if a subnet is connected to an internet gateway, the subnet is public.

Attaching an internet gateway to a VPC means that resources in my VPC can now access the internet and instances of EC2 with public IP addresses also become accessible to users therefore, applocations hosted on those servers become public too. If I missed this step, my VPC would not be able to connect to the internet.





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